15/04/2025

CODE-A

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

CST-5

Time : 180 Mins.

Answers

1. (2)	37. (4)	73. (2)	109. (4)	145. (4)
2. (2)	38. (3)	74. (2)	110. (2)	146. (3)
3. (4)	39. (1)	75. (3)	111. (3)	147. (1)
4. (1)	40. (3)	76. (1)	112. (2)	148. (1)
5. (3)	41. (1)	77. (2)	113. (3)	149. (2)
6. (4)	42. (4)	78. (1)	114. (2)	150. (1)
7. (3)	43. (3)	79. (3)	115. (4)	151. (3)
8. (4)	44. (3)	80. (1)	116. (3)	152. (3)
9. (4)	45. (3)	81. (1)	117. (4)	153. (1)
10. (3)	46. (1)	82. (4)	118. (1)	154. (4)
11. (2)	47. (3)	83. (2)	119. (2)	155. (2)
12. (2)	48. (4)	84. (1)	120. (2)	156. (2)
13. (1)	49. (4)	85. (3)	121. (3)	157. (3)
14. (3)	50. (2)	86. (3)	122. (3)	158. (2)
15. (4)	51. (2)	87. (1)	123. (1)	159. (2)
16. (1)	52. (4)	88. (3)	124. (3)	160. (2)
17. (2)	53. (4)	89. (4)	125. (3)	161. (3)
18. (4)	54. (3)	90. (4)	126. (2)	162. (3)
19. (4)	55. (1)	91. (3)	127. (3)	163. (1)
20. (3)	56. (2)	92. (1)	128. (3)	164. (4)
21. (4)	57. (1)	93. (3)	129. (3)	165. (2)
22. (4)	58. (4)	94. (3)	130. (3)	166. (2)
23. (2)	59. (1)	95. (2)	131. (4)	167. (2)
24. (1)	60. (4)	96. (3)	132. (2)	168. (4)
25. (3)	61. (3)	97. (3)	133. (1)	169. (2)
26. (2)	62. (2)	98. (2)	134. (4)	170. (2)
27. (1)	63. (4)	99. (2)	135. (4)	171. (2)
28. (1)	64. (1)	100. (3)	136. (2)	172. (2)
29. (1)	65. (1)	101. (1)	137. (1)	173. (2)
30. (2)	66. (3)	102. (3)	138. (1)	174. (4)
31. (1)	67. (1)	103. (3)	139. (1)	175. (3)
32. (1)	68. (4)	104. (4)	140. (2)	176. (1)
33. (2)	69. (3)	105. (4)	141. (2)	177. (2)
34. (4)	70. (1)	106. (4)	142. (3)	178. (1)
35. (1)	71. (3)	107. (1)	143. (4)	179. (2)
36. (4)	72. (3)	108. (1)	144. (3)	180. (3)

Hints and Solutions

BIOLOGY

(1) **Answer :** (2)
Solution:
Properties of tissue arise as a result of interactions among the constituent cells

(2) **Answer :** (2)
Solution:
All viruses have protein and genetic material, that can be either DNA or RNA. Viruses are inert outside the host.

(3) **Answer :** (4)
Solution:
Presence of plasmodium (i.e. naked body) is an animal like feature of slime mould.

(4) **Answer :** (1)
Solution:
Albugo causes white spots on mustard leaves. *Albugo* is a member of phycomycetes and it lacks dikaryophase in its life.

(5) **Answer :** (3)
Solution:
Antheridium and archegonium are male and female sex organs respectively.

(6) **Answer :** (4)
Solution:
Pyrenoids are present in green algae. They store starch.

(7) **Answer :** (3)
Solution:
Rose has half-inferior ovary with apocarpous condition. Mustard has superior ovary in syncarpous condition. Guava and sunflower have inferior ovary.

(8) **Answer :** (4)
Solution:
Colchicine is derived from *Colchicum autumnale*. It is used to arrest cell division. *Colchicum autumnale* belongs to family Liliaceae.

(9) **Answer :** (4)
Solution:
Meristematic cells are thin walled with dense protoplasm

(10) **Answer :** (3)
Solution:
Staminode – Sterile stamen
Syncarpous – Fused carpels
Gamopetalous – Petals united
Polysepalous – Sepals free

(11) **Answer :** (2)
Solution:
Intercalary meristem helps in the elongation of the plant organ. This tissue is intercalated between mature tissues

(12) **Answer :** (2)
Solution:
Mesophyll cells can perform photosynthesis. Monocot stem has sclerenchymatous hypodermis.

(13) **Answer :** (1)
Solution:
Outer side of epidermis is often covered by cuticle which is absent in roots.

(14) **Answer :** (3)
Solution:
Polar heads are directed towards the cytoplasm and non polar tails towards the inner side. In this way polar heads face the aqueous environment and non polar tails are protected from aqueous environment.

(15) **Answer :** (4)

Solution:
RER synthesizes proteins which are secreted outside the cell and for golgi, lysosomes, plasma membrane and ER itself.

(16) Answer : (1)
Solution:
Cells, tissues, organs and organ systems split up the work in a way that exhibits division of labour and contribute to survival of the body as a whole. Protozoans, being unicellular, do not show division of labour and single cell has the ability to perform all of the functions required for their survival.

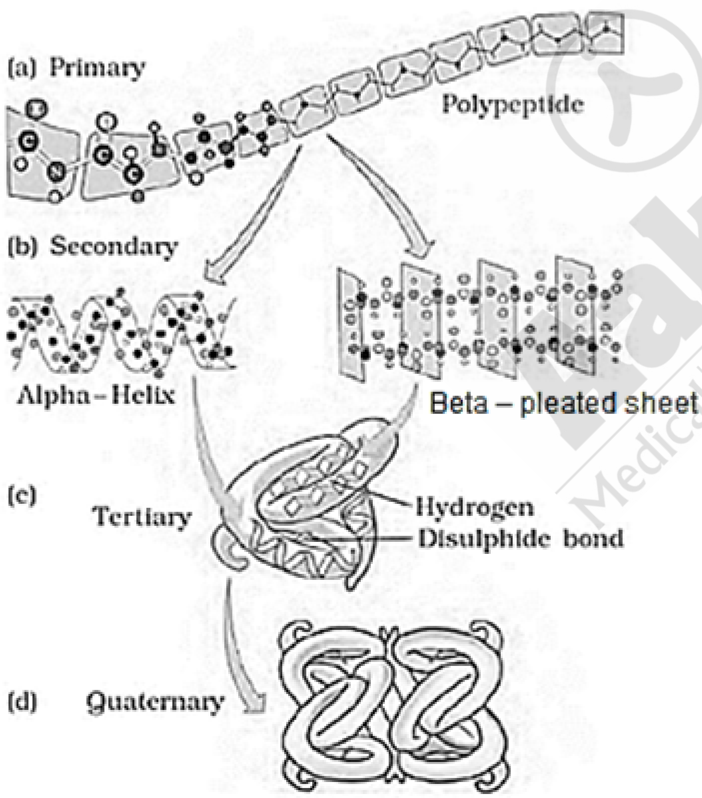
(17) Answer : (2)
Solution:
Glutamic acid and aspartatic acid are acidic amino acids.
Uridylic acid and thymidylic acid are nucleotides. Palmitic acid is a fatty acid.

(18) Answer : (4)
Solution:
Compound epithelium covers the dry surface of pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

(19) Answer : (4)
Solution:
Cockroach has a network of tubes (tracheal tubes) to transport atmospheric air within the body.

(20) Answer : (3)
Solution:
Parasympathetic neural signals decrease the rate of heart beat.

(21) Answer : (4)
Solution:
In a polypeptide chain, amino acids are joined together by peptide bonds.

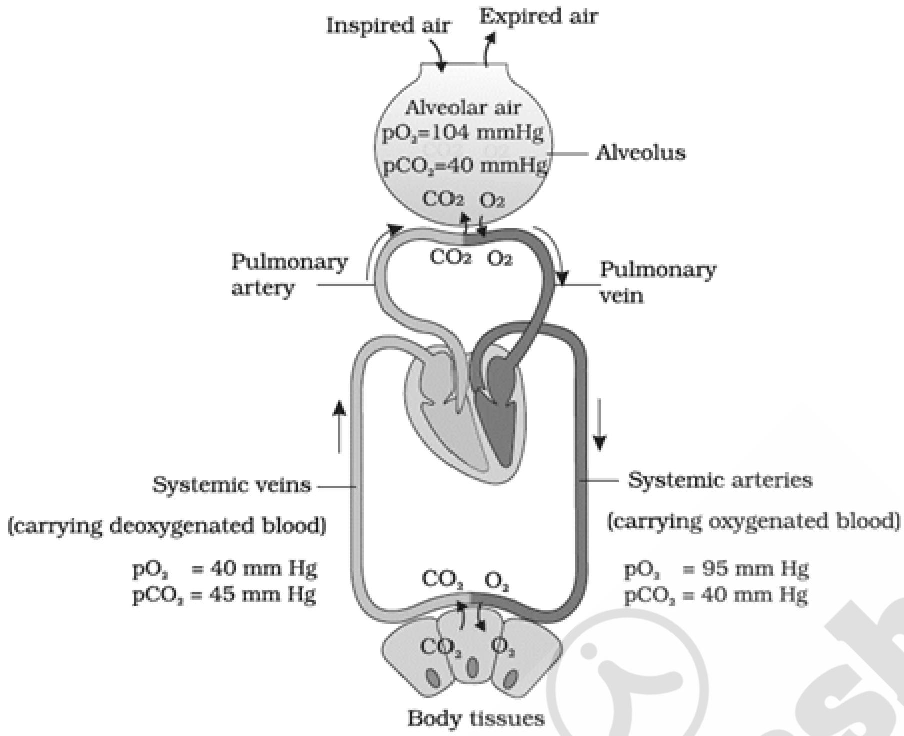


(22) Answer : (4)
Solution:
Disulphide bond is present in tertiary and quaternary structures of proteins such as pro-insulin, insulin and antibodies.
Glycosidic bonds are present between monosaccharides in glycogen.

(23) Answer : (2)
Solution:
Mechanisms of breathing vary among different groups of animals depending mainly on their habitats and levels of organisation.

(24) Answer : (1)
Solution:
Glycine is unique in being optically inactive as it has hydrogen as the R group. It is the simplest amino acid.

(25) Answer : (3)
Solution:



(26) Answer : (2)
Solution:
Chlorophyll pigments are present in thylakoids.

(27) Answer : (1)
Solution:
Duplication of centrioles occurs during S phase of the interphase. The most dramatic phase of cell cycle is M phase.

(28) Answer : (1)
Solution:
Spindle fibres begins to form during prophase and they attach to the kinetochore of the chromosomes during metaphase.

(29) Answer : (1)
Solution:
The sequence of cell cycle is $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
After M-phase the cell will entre again in G_1 phase. G_0 stage cell re-entre G_1 phase.

(30) Answer : (2)
Solution:
NADP reductase is located on the stroma side of thylakoid membrane
In Calvin cycle, reduction require 2ATP and 2NADPH

(31) Answer : (1)
Solution:
Crossing over occurs during prophase I.
Homologous chromosomes separate during anaphase I

(32) Answer : (1)
Solution:
In ETS, the energy of oxidation-reduction is utilized to create the proton gradient in mitochondria.

(33) Answer : (2)
Solution:

The reactions catalysed by pyruvic dehydrogenase require the participation of several coenzymes, including NAD^+ and coenzyme A

(34) **Answer :** (4)
Solution:
Autoclaved herring sperm DNA, corn kernel suspension and coconut milk are the sources for the cytokinin. Cytokinin promotes nutrient mobilization which helps in the delay of leaf senescence.

(35) **Answer :** (1)
Solution:
Coconut milk contain cytokinin, it promotes cell division.

(36) **Answer :** (4)
Solution:
Endosperm development precedes embryo development. Early stages of embryogeny is same in both monocots and dicots.

(37) **Answer :** (4)
Solution:
The pollen grains in angiosperm are called male gametophyte.

(38) **Answer :** (3)
Solution:
Thalassemia is related to autosomal genes and it is a quantitative problem of synthesising globin molecule. Since, the child in the given case will have two mutated alleles out of the four, he/she will show mild symptom of the disorder.

(39) **Answer :** (1)
Solution:
Recombination frequency is used to prepare genetic maps. Recombination frequency of closely situated genes is low.

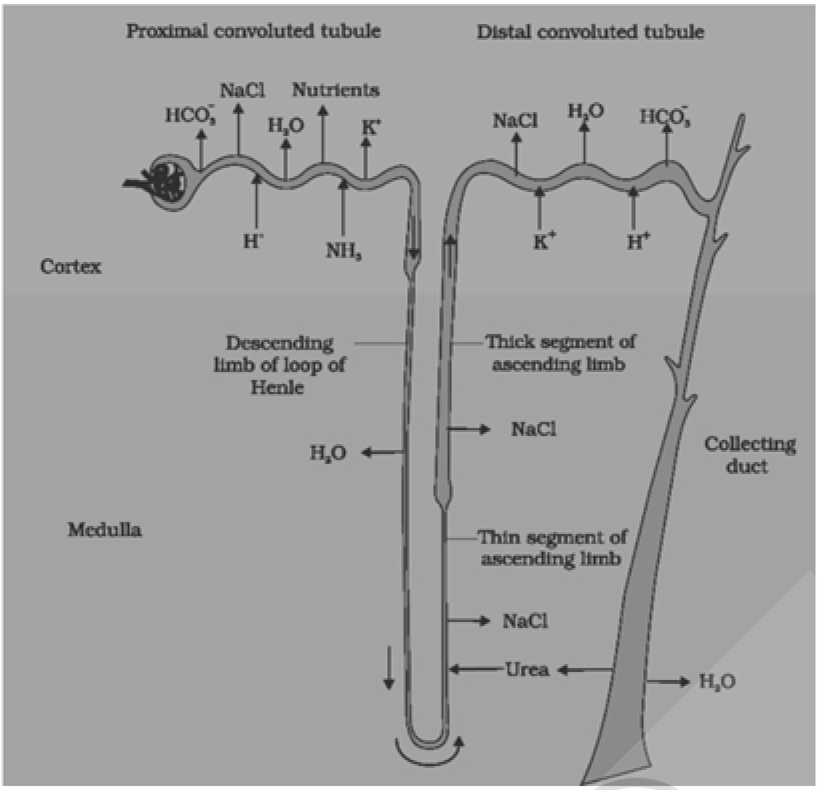
(40) **Answer :** (3)
Solution:
Myotonic dystrophy is a dominant autosomal trait. This trait is inherited in the children only if both or any one of the parents shows the trait.

(41) **Answer :** (1)
Solution:
Scapula is a large triangular bone of shoulder which possesses cavity known as glenoid cavity that articulates with head of humerus.

(42) **Answer :** (4)
Solution:
The heart, like any other muscle, becomes stronger through regular exercise. A stronger heart is expected to have a greater stroke volume, which would lead to decrease in heart rate.

(43) **Answer :** (3)
Solution:
During impulse conduction through a nerve fibre, current flows towards axon terminals on inner side of axonal membrane and towards cyton on outer side of axonal membrane to complete the circuit of current flow.

(44) **Answer :** (3)
Solution:
Sodium is actively reabsorbed in PCT and DCT but passively reabsorbed in thin segment of ascending limb of Henle's loop.



(45) **Answer :** (3)
Solution:
Thymosin is secreted by thymus gland which is located in the mediastinum of lungs.

(46) **Answer :** (1)
Solution:
Coelenterates are diploblastic acoelomates whereas platyhelminths are triploblastic acoelomates. Hookworm is a triploblastic pseudocoelomate.

(47) **Answer :** (3)
Solution:
CNS is the site for information processing and control which includes both brain and spinal cord.

(48) **Answer :** (4)
Solution:
The Government of India legalised MTP with some strict conditions to avoid its misuse. Such restrictions are important to check indiscriminate and illegal female foeticide.

(49) **Answer :** (4)
Solution:
Triceratops was a dinosaur with a three-horned face. It was a herbivore and quadriped.

(50) **Answer :** (2)
Solution:
In male frogs, ureters act as the urinogenital duct that carries both sperms and urine. Whereas in females, there is no functional connection between ovaries and kidneys.
Thus, if ureters of male frogs are cut, gamete transport will get affected.

(51) **Answer :** (2)
Solution:
Mendel had talked of inheritable ‘factors’ influencing phenotype. Saltation is single step large mutation. Hugo de Vries believed mutation caused speciation and not the minor variations that Darwin talked about.

(52) **Answer :** (4)
Solution:
ADH released from neurohypophysis facilitates water reabsorption from DCT of nephron and also shows its constrictory effect on blood vessels.

(53) **Answer :** (4)
Solution:

If n = Number of restriction sites in a DNA, then the number of fragments obtained from a linear DNA after restriction digestion = n + 1.
While for circular DNA, it is equal to n.

(54) Answer : (3)
Solution:

Thomas Malthus	–	His work on populations influenced Darwin
Louis Pasteur	–	Theory of biogenesis
Ernst Haeckel	–	Proposed embryological support for evolution
Alfred Wallace	–	Malay Archipelago
Darwin	–	H.M.S. Beagle

(55) Answer : (1)
Solution:
Typhoid is characterised by sustained high fever (39°C to 40°C), weakness, stomach pain, constipation, headache and loss of appetite.
Bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for the disease pneumonia in humans which infects the alveoli of the lungs.

(56) Answer : (2)
Solution:
The electrical activity of the heart can be recorded from body surface by using electrocardiograph and the recorded graph is called electrocardiogram. Pons is absent in frogs.

(57) Answer : (1)
Solution:
Traditional hybridisation procedures used in plant and animal breeding often lead to inclusion and multiplication of undesirable genes along with the desired genes.

(58) Answer : (4)
Solution:
Muscle contains a red coloured oxygen storing pigment called myoglobin. Myoglobin content is high in some of the muscles which gives a reddish appearance. Such muscles are called the red fibres. These muscles also contain plenty of mitochondria which are used for ATP production. Contractility in both white and red muscle fibres is due to presence of both actin and myosin filaments.

(59) Answer : (1)
Solution:
Cervical caps are barriers made up of rubber that are inserted into the female reproductive tract to cover the cervix during coitus.

(60) Answer : (4)
Solution:
Heart is usually three-chambered in reptiles but in crocodiles it is four-chambered.
In birds and mammals, heart is four-chambered.

(61) Answer : (3)
Solution:
Biology states that mind influences through neural system and endocrine system, our immune system and that our immune system maintains our health. Hence, mind and mental state can affect our health.

(62) Answer : (2)
Solution:
Vasopressin acts mainly at the kidney and stimulates reabsorption of water and electrolytes by the distal tubules and thereby reduces loss of water through urine (diuresis). Hence, it is also called antidiuretic hormone (ADH).

Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions. Thus, aldosterone helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.
Thyroid hormones play an important role in the regulation of the basal metabolic rate. These hormones also support the process of red blood cell formation. Thyroid hormones control the metabolism of carbohydrates, proteins and fats. Maintenance of water and electrolyte balance is also influenced by thyroid hormones.

(63) Answer : (4)
Solution:

Transgenic animals are made to carry genes which make them more sensitive to toxic substances than non-transgenic animals.

(64) Answer : (1)
Solution:
Notochord is a mesodermally derived rod-like structure formed on the dorsal side during embryonic development in some animals. Animals with notochord are called chordates and those animals which do not form this structure are called non-chordates, *e.g.,* porifers to echinoderms.

(65) Answer : (1)
Solution:
The recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on tetracycline containing medium.

(66) Answer : (3)
Solution:
Turner's syndrome is caused due to the absence of one of the X-chromosomes i.e, 45 with XO.

(67) Answer : (1)
Solution:
X-chromosome was termed after the discovery of X-body by Henking.

(68) Answer : (4)
Solution:
Sulphur is the component of proteins. The capsid of the bacteriophage is formed within the bacteria by its translation machinery.

(69) Answer : (3)
Solution:
An mRNA has many non-coding stretches of nucleotides and the codons are always triplet. Therefore, if there are n number of nucleotides in an mRNA, the maximum number of amino acids in the polypeptide synthesised from this mRNA would be approximately $\frac{n}{3}$.

(70) Answer : (1)
Solution:
Regulator gene in *lac* operon codes for repressor protein.

(71) Answer : (3)
Solution:
Friedrich Meischer named the acidic substance present in nucleus as Nuclein.

(72) Answer : (3)
Solution:
23S rRNA in bacteria is the ribozyme

(73) Answer : (2)
Solution:
The blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequence and later assigning different regions in the sequence with functions is termed sequence annotation

(74) Answer : (2)
Solution:
A-secondary effluent, B-river water, C-primary effluent and D-untreated sewage water.

(75) Answer : (3)
Solution:
Oscillatoria can fix atmospheric nitrogen in an aquatic environment

(76) Answer : (1)
Solution:
Calotropis produces cardiac glycosides.

(77) Answer : (2)
Solution:
Majority of animals and all plants are conformers.

(78) Answer : (1)
Solution:

Plants	→	Caterpillar (10,000 J)	→	Sparrow (1000 J)	→	Hawk

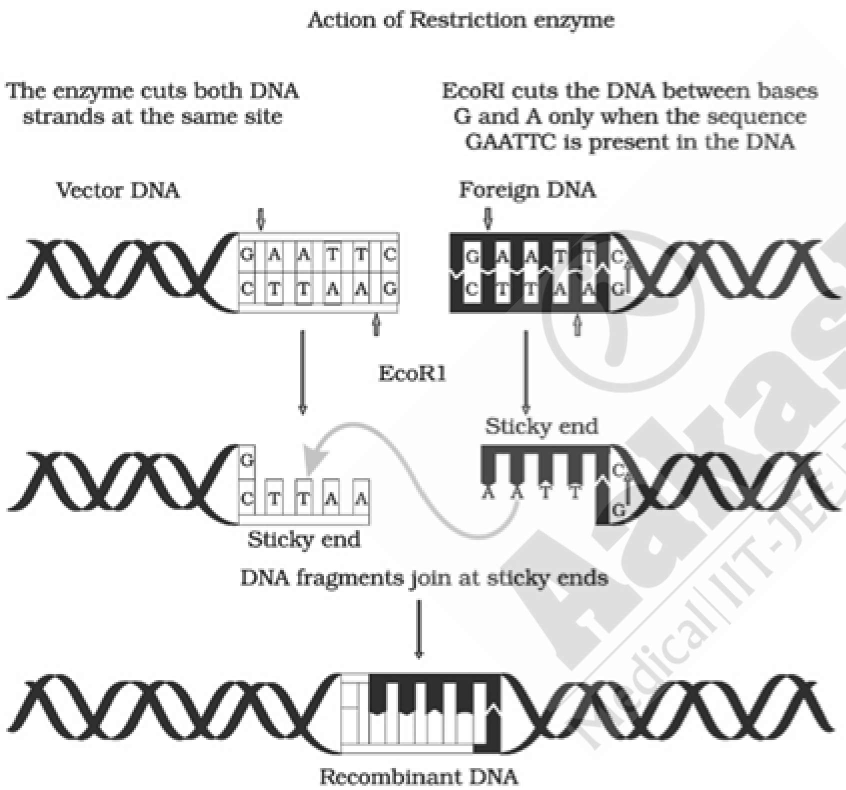
(100,000 J)						(100 J)
T_1		T_2		T_3		T_4

(79) **Answer :** (3)
Solution:
Value of z lies in the range of 0.1 to 0.2 regardless of the taxonomic group or the region.

(80) **Answer :** (1)
Solution:
A stable community should not show too much variation in productivity from year to year; it must be either resistant or resilient to occasional disturbances and it must also be resistant to invasions by alien species.

(81) **Answer :** (1)
Solution:
Using *Agrobacterium* vectors, nematode-specific genes were introduced into the host plant. The introduction of DNA was such that it produced both sense and anti-sense RNA in the host cells.

(82) **Answer :** (4)
Solution:



(83) **Answer :** (2)
Solution:

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(84) **Answer :** (1)
Solution:
The blastomeres in the blastocyst are arranged into an outer layer called trophoblast and an inner group of cells attached to trophoblast called the inner cell mass. The trophoblast layer then gets attached to the endometrium and the inner cell mass gets differentiated as the embryo. After attachment, the uterine cells divide rapidly and covers the blastocyst. As a result, the blastocyst becomes embedded in the endometrium of the uterus.

(85) Answer : (3)
Solution:
The convention for naming restriction enzymes is that the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, *e.g.*, *Eco*RI comes from *Escherichia coli* RY 13. In *Eco*RI, the letter 'R' is derived from the name of strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria.

(86) Answer : (3)
Solution:
Specific Bt toxin genes were isolated from *Bacillus thuringiensis* and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group specific. The toxin is coded by a gene named *cry*. There are a number of them, for example, the proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

(87) Answer : (1)
Solution:
Intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy (MTP) or induced abortion. Nearly 45 to 50 million MTPs are performed in a year all over the world which accounts to 1/5th of the total number of conceived pregnancies in a year.

(88) Answer : (3)
Solution:
Each peak in the ECG is identified with a letter from P to T that corresponds to a specific electrical activity of the heart. The P-wave represents the electrical excitation (or depolarisation) of the atria, which leads to the contraction of both the atria. The QRS complex represents the depolarisation of the ventricles, which initiates the ventricular contraction. The contraction starts shortly after Q and marks the beginning of the systole. The T-wave represents the return of the ventricles from excited to normal state (repolarisation). The end of the T-wave marks the end of systole.

(89) Answer : (4)
Solution:
The opening of the vagina is often covered partially by a membrane called hymen. The hymen is often torn during the first coitus (intercourse). However, it can also be broken by a sudden fall or jolt, insertion of a vaginal tampon, active participation in some sports like horseback riding, cycling, *etc.* In some women, the hymen persists even after coitus. In fact, the presence or absence of hymen is not a reliable indicator of virginity or sexual experience.

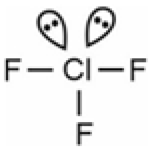
(90) Answer : (4)
Solution:
Placenta also acts as an endocrine tissue and produces several hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progestogens, *etc*. In the later phase of pregnancy, a hormone called relaxin is also secreted by the ovary.

CHEMISTRY

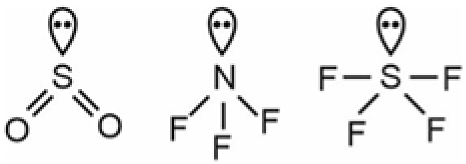
(91) Answer : (3)
Solution:
Group 17 elements (halogens) have very high negative electron gain enthalpies because they can attain stable noble gas electronic configuration by picking up an electron.
However, electron gain enthalpy is more negative in case of third period element than second period element as the added electron goes to third shell which suffers much less electron-electron repulsion when if goes to second shell

(92) Answer : (1)
Solution:
It is not necessary for a polyatomic molecule to be linear to show zero dipole moment, molecules like CH_4, BF_3 etc are not linear but have zero dipole moment.

(93) Answer : (3)
Solution:



(T-shape)



(Bent) (Trigonal pyramidal) (See-saw)

(94) Answer : (3)

Solution:

(1)

	CH_3COOH +	$NaOH$ →	$CH_3COONa + H_2O$
Initial Milliequivalent	50 × 0.1 = 5 meq.	50 × 0.1 5 meq.	
	—	—	5 meq.

• Not a buffer solution.

(2)

	$NaOH$ +	HCl →	$NaCl + H_2O$
Initial Milliequivalent	50 × 0.1 = 5 meq.	100 × 0.1 = 10 meq.	
	—	5 meq.	5 meq.

• Not a buffer solution.

(3)

	CH_3COONa +	HCl →	+ $CH_3COOH + NaCl$
Initial Milliequivalent	100 × 0.1 = 10 meq.	50 × 0.1 = 5 meq.	
	5 meq.	—	5 meq.

• Mixture of ($CH_3COONa + CH_3COOH$) is a acidic buffer.

(4)

	CH_3COOH +	$NaOH$ →	+ $CH_3COONa + H_2O$
Initial Milliequivalent	25 × 0.1 = 2.5 meq.	50 × 0.1 = 5 meq.	
	—	2.5 meq.	2.5 meq.

• Not a buffer solution.

(95) Answer : (2)

Solution:

For same type of salts, as K_{sp} increases, solubility of salt also increases.

- K_{sp} order is, $CuCl > PbCO_3 > SnS$
- Solubility order will be, $CuCl > PbCO_3 > SnS$

(96) Answer : (3)

Solution:

Step 1: The ionic equation is

$$MnO_4^{\ominus}(aq) + Br^{\ominus}(aq) \rightarrow MnO_2(s) + BrO_3^{\ominus}(aq)$$

Step 2: Assign oxidation number of Mn and Br and make the increase equal to decrease in oxidation number.

$$2\overset{+7}{\underline{Mn}}O_4^{\ominus}(aq) + \underline{Br}^{\ominus}(aq) \rightarrow 2\overset{+4}{\underline{Mn}}O_2(s) + \overset{+5}{\underline{Br}}O_3^{\ominus}(aq)$$

Step 3: As the reaction occurs in basic medium, add OH^- ions and H_2O appropriately.

$$2MnO_4^{\ominus}(aq) + Br^{-}(aq) + H_2O(l) \rightarrow 2MnO_2(s)$$

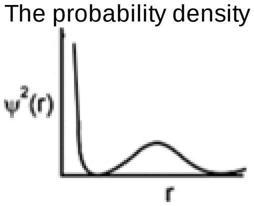
$$+ BrO_3^{\ominus}(aq) + 2OH^{-}(aq)$$

(97) **Answer :** (3)
Solution:

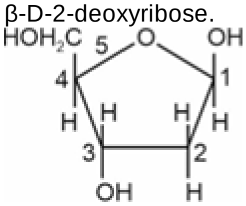
$$\overset{0}{\underline{Cl}_2}(g) + 2NaOH(aq) \rightarrow NaO\overset{+1}{\underline{Cl}}(aq) + Na\overset{-1}{\underline{Cl}}(aq) + H_2O(l)$$

Here the chlorine of Cl_2, which is present in zero oxidation state, is converted to +1 oxidation state in NaOCl and decreases to –1 oxidation state in NaCl. So, chlorine undergoes disproportionation reaction with NaOH.

(98) **Answer :** (2)
Solution:
The probability density distribution curve for 2s orbital is



(99) **Answer :** (2)
Solution:
Sugar moiety present in DNA is
β-D-2-deoxyribose.

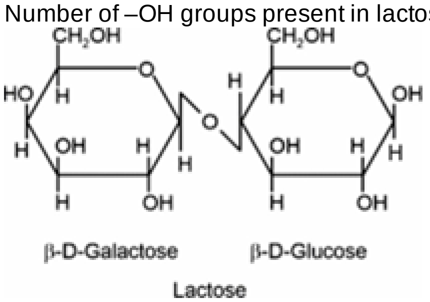


(100) **Answer :** (3)
Solution:
Total energy = nε
n → no of photon ε → energy of one photon

$$\therefore\ n \times \frac{hc}{\lambda} = 1 \quad \left[\varepsilon = h\nu = h\frac{c}{\lambda}\right]$$

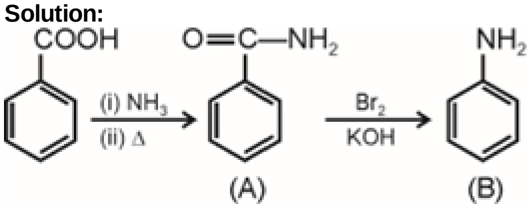
$$n = \frac{1 \times \lambda}{h \times c} = \frac{1 \times 5000 \times 10^{-10}}{6.63 \times 10^{-34} \times 3 \times 10^{8}} = 2.5 \times 10^{18}$$

(101) **Answer :** (1)
Solution:
Number of –OH groups present in lactose are 8.



(102) **Answer :** (3)
Solution:
Electron withdrawing group (–NO_2) decreases basic strength of the compound. pK_b of p-nitroaniline will be highest.

(103) **Answer :** (3)
Solution:



(104) **Answer :** (4)

Solution:

$$CH_3\,CH_2-OH \xrightarrow{SOCl_2} CH_3\,\underset{A}{CH_2}-Cl \xrightarrow{KCN}$$

$$CH_3\,\underset{B}{CH_2}-CN \xrightarrow[(ii)\ H_2O]{(i)\ DIBAL-H} CH_3\,\underset{C}{CH_2}\,CHO$$

Product (C) undergoes aldol condensation reaction in alkaline medium.

(105) Answer : (4)

Solution:

$K = \left(\frac{2.303}{t}\right)\left(\log\frac{P_i}{P_A}\right)$

		A(g)	$\longrightarrow$	B(g)	+	2C(g)
At	$t = 0$	p_i		0		0
At	time (t)	$p_i - x$		x		2x

$P_t = (P_i - x) + x + 2x = P_i + 2x$

$x = \frac{P_t - P_i}{2}$

$P_A = P_i - x = P_i - \frac{P_t - P_i}{2} = \frac{3P_i - P_t}{2}$

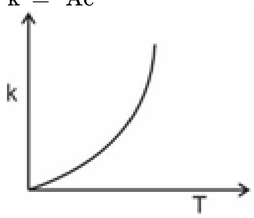
so $K = \left(\frac{2.303}{t}\right)\left(\log\frac{2P_i}{3P_i - P_t}\right)$

(106) Answer : (4)

Solution:

The Arrhenius equation showing variation of rate constant vs temperature

$k = Ae^{-E_a/RT}$



(107) Answer : (1)

Solution:

$$CH_3\,CH_2\,OH \xrightarrow[573K]{Cu} CH_3\,CHO\left(\text{Major}\right)$$

(108) Answer : (1)

Solution:

68% nitric acid and 32% water by mass behaves as maximum boiling azeotrope.

(109) Answer : (4)

Solution:

$$p_{N_2} = K_H \times \left(\frac{n_{N_2}}{n_{N_2} + n_{H_2O}}\right)_{\text{solution}} \simeq K_H \times \frac{n_{N_2}}{n_{H_2O}}$$

$$5 \times \frac{4}{5} = 10^5 \times \frac{n_{N_2}}{2} \quad \left[\because\ p_{N_2} = p_T\,x_{N_2}\right]$$

$$n_{N_2} = 8 \times 10^{-5}$$

(110) Answer : (2)

Solution:

Average atomic mass $= \frac{10+80+15\times 20}{100}$

$= \frac{800+300}{100}$

= 11 u

(111) Answer : (3)
Solution:
10^{15} is called peta

(112) Answer : (2)
Solution:
$\frac{1}{2}\mathrm{H_2} + \frac{1}{2}\mathrm{Cl_2}(\mathrm{g}) \rightarrow \mathrm{HCl}(\mathrm{g})$

$\Delta \mathrm{H_f} = \frac{1}{2}(434 + 242) - 431$

= –93 kJ mol^{-1}

(113) Answer : (3)
Solution:
$\mathrm{N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g); K_c}$

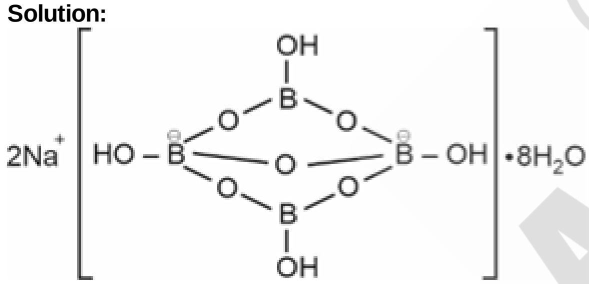
$2\,\mathrm{NH_3}\left(\mathrm{g}\right) \rightleftharpoons \mathrm{N_2}\left(\mathrm{g}\right) + 3\mathrm{H_2}\left(\mathrm{g}\right); \frac{1}{\mathrm{K_c}}$

$\mathrm{NH_3}\left(\mathrm{g}\right) \rightleftharpoons \frac{1}{2}\mathrm{N_2}\left(\mathrm{g}\right) + \frac{3}{2}\mathrm{H_2}\left(\mathrm{g}\right); \frac{1}{\sqrt{\mathrm{K_c}}}$

(114) Answer : (2)
Solution:
$$\mathrm{Ni\,(OH)_2} \rightleftharpoons \underset{(\mathrm{s})}{\mathrm{Ni^{2+}}}(\mathrm{aq}) + 2\,\underset{\underset{\simeq 0.2}{(0.2+\mathrm{s})}}{\mathrm{OH^-}}(\mathrm{aq})$$

$s(0.2)^2 = 2 \times 10^{-15}$
$s = 5 \times 10^{-14}$

(115) Answer : (4)
Solution:



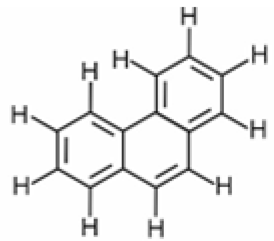
(116) Answer : (3)
Solution:
Hyperconjugation involves delocalisation of σ electrons of C – H bond of an alkyl group directly attached to an atom of unsaturated system or to are atom with an unshared p-orbital.

(117) Answer : (4)
Solution:
233 g of $BaSO_4$ contains 32g of S

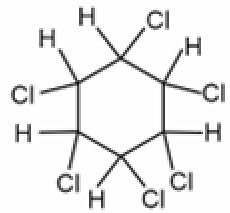
0.466 g of $BaSO_4$ contains $\frac{32 \times 0.466}{233}$ g of S

% of sulphur = $\frac{32 \times 0.466 \times 100}{233 \times 0.25} = 25.6\%$

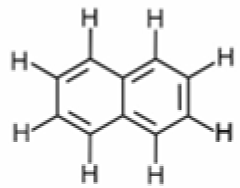
(118) Answer : (1)
Solution:



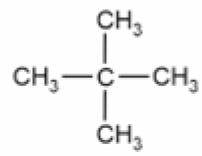
Phenanthrene
(Number of σ bonds = 26)



Gammaxane
(Number of σ bonds = 18)



Naphthalene
(Number of σ bonds = 19)



Neopentane
(Number of σ bonds = 16)

(119) Answer : (2)
Solution:

$$\mathrm{CH_3\,COO^-\,Na^+ + NaOH \xrightarrow[\Delta]{CaO} CH_4 + Na_2\,CO_3}$$

(120) Answer : (2)
Solution:
At high altitude, the total pressure decreases hence partial pressure of oxygen also decreases.

(121) Answer : (3)
Solution:

$$\mathrm{Ni(s) + 2\,Ag^+(0.002M) \rightarrow 2\,Ag(s) + Ni^{2+}(0.16M)}$$

$$E_{cell}^{\circ} = E_{cell}^{\circ} - \frac{0.059}{2}\log\frac{[Ni^{2+}]}{[Ag^+]^2}$$

$$= 1.05 - \frac{0.059}{2}\log\frac{0.16}{(0.002)^2}$$

(122) Answer : (3)
Solution:

Compound	$\Delta_{diss}H^o$(kJ mol^{-1})
HF	574
HCl	432
HBr	363
HI	295

(123) Answer : (1)
Solution:
Ionisation of an atom is always an endothermic process.

(124) Answer : (3)
Solution:

$$3\mathrm{H\overset{+3}{N}O_2} \rightarrow \mathrm{H\overset{+5}{N}O_3 + H_2O + 2\,NO}\left(\text{Nitric oxide}\right)$$

(125) Answer : (3)
Solution:

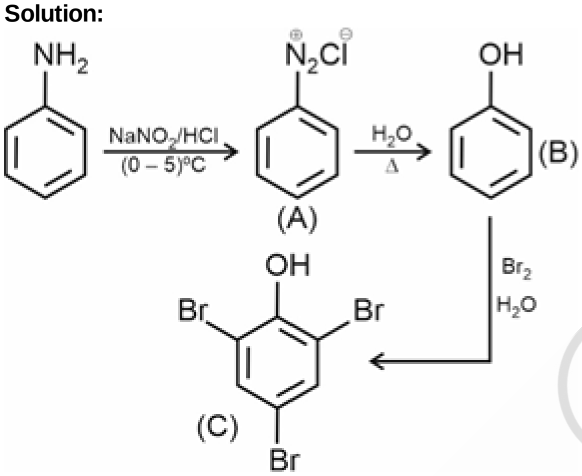
Ion	Radii (pm)
La^{3+}	106
Eu^{3+}	95
	86

Yb^{3+}	

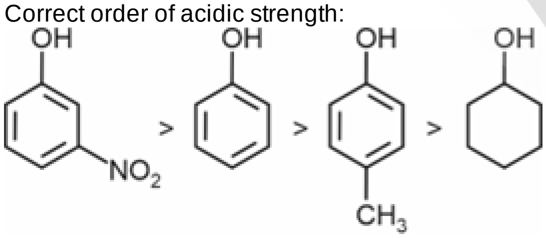
(126) Answer : (2)
Solution:
Both rhombic and monoclinic sulphur have S_8 molecules. These S_8 molecules are packed to give different crystal structures.

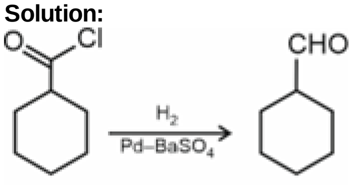
(127) Answer : (3)
Solution:
M – C π bond is formed by the donation of a pair of electrons from a filled *d*-orbital of metal into the vacant antibonding π* orbital of CO.

(128) Answer : (3)
Solution:
Propan-2-ol does not contain chiral centre.

(129) Answer : (3)
Solution:


(130) Answer : (3)
Solution:
Ethanol is polar protic solvent.

(131) Answer : (4)
Solution:
Correct order of acidic strength:


(132) Answer : (2)
Solution:


Rosenmund reduction

(133) Answer : (1)
Solution:
$$SO_2 + Cl_2 \xrightarrow{\text{Charcoal}} \underset{\text{Suphuryl chloride}}{SO_2\,Cl_2}$$

(134) Answer : (4)

Solution:
More stable is the carbocation, faster is S_N1 rate.

(135) Answer : (4)
Solution:

Species	Number of unpaired electrons
$[Fe(CN)_6]^{3-}$	1
$[Co(H_2O)_6]^{3+}$	Zero
$[FeF_6]^{3-}$	5
$[NiCl_4]^{2-}$	2

PHYSICS

(136) Answer : (2)
Solution:

$$\begin{array}{r} 436.32 \text{ g} \\ 227.2 \text{ g} \\ +\ 0.301 \text{ g} \\ \hline 663.821 \text{ g} \\ \hline \end{array}$$

in 227.2 only one digit is after decimal.
So, final result is 663.8 g.

(137) Answer : (1)
Solution:
Displacement in nth second is given by
$S_{nth} = u + \frac{a}{2}(2n-1)$
$20 = u + \frac{a}{2}(2 \times 4 - 1)$
$20 = u + \frac{a}{2} \times 7$ ...(i)
$25 = u + \frac{a}{2} \times 9$ ...(ii)
Equation (ii) –equation (i)
$25 - 20 = \frac{9a}{2} - \frac{7a}{2}$
Putting this in equation (i), we get
$20 = u + \frac{35}{2}$
$u = \frac{5}{2}$
Distance in 3rd second
$S = u + \frac{a}{2}(2 \times 3 - 1)$
$S = u + \frac{a}{2} \times 5$
$S = \frac{5}{2} + \frac{5}{2} \times 5$
$S = \frac{30}{2} = 15\text{m}$

(138) Answer : (1)
Solution:
For stone 1 : $x_1 = 10t - \frac{1}{2}gt^2$ ...(i)
For stone 2 : $x_2 = 20t - \frac{1}{2}gt^2$ ...(ii)
Relative position of (2) w.r.t (1)
$x_2 - x_1 = 20t - \frac{1}{2}gt^2 - \left(10t - \frac{1}{2}gt^2\right)$
$x_2 - x_1 = 10t$

∴ The grape will be a straight line

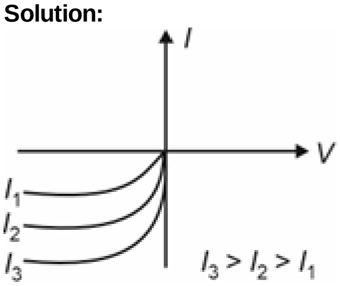
(139) **Answer :** (1)
Solution:
If velocity of one particle w.r.t other particle is along the line joining them, then only the two particles can meet.

(140) **Answer :** (2)
Solution:
Equation of trajectory of a projectile:
$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$
Given $y = \sqrt{3}x - 5x^2$
On comparing, we get
$\tan\theta = \sqrt{3} \Rightarrow \theta = 60^{\circ}$ from horizontal.
The angle of projection with vertical is
$\alpha = 90^{\circ} - \theta \Rightarrow \alpha = 30^{\circ}$

(141) **Answer :** (2)
Solution:
$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$
$= \left(\frac{8-4}{8+4}\right)g$
$= \frac{g}{3}$

(142) **Answer :** (3)
Solution:
Minimum potential energy corresponds to stable equilibrium.

(143) **Answer :** (4)
Solution:
According to work-energy theorem, net work done is equal to change in kinetic energy of a system

(144) **Answer :** (3)
Solution:


As light intensity increases, reverse bias current increases. Because in reverse bias, current is due to minority carrier and as intensity increases, current variation can be observed easily

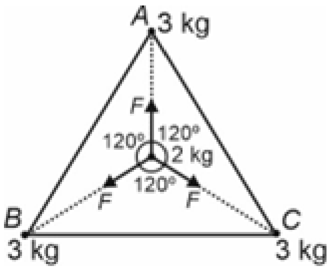
(145) **Answer :** (4)
Solution:
The Zener diode in the circuit will regulate the voltage and hence V_0 = 6V

(146) **Answer :** (3)
Solution:
Nuclear density is independent of mass number. Binding energy per nucleon is maximum for Iron (A = 56).
Neutrons and protons are bound in a nucleus by strong nuclear force.

(147) **Answer :** (1)
Solution:
$^{238}_{z}X \rightarrow {}^{234}_{z-2}Y + \underset{\alpha-\text{particle}}{^{4}_{2}He}$
$K.E. = \frac{P^2}{2m}$ and in this reaction total momentum remains conserved.
5.4 × 4 = KE_y × 234
$KE_y = \frac{5.4\times4}{234} = 0.092\ \text{MeV}$
Q-value = total kinetic energy
Q-value = 5.4 + .092 = 5.49 MeV

(148) Answer : (1)
Solution:
Time period of electron in a hydrogen atom
$T \propto n^3$
$\frac{8T}{T} = \frac{n_1^3}{n_2^3} \Rightarrow n_1 = 2n_2$
This is satisfied by n_2 = 2 and n_1 = 4

(149) Answer : (2)
Solution:
$\vec{r}_{CM} = \frac{\left(\hat{i}+\hat{j}+\hat{k}\right)\times 2+\left(2\hat{i}+2\hat{j}+\hat{k}\right)6}{8}$
$= \frac{14\hat{i}+14\hat{j}+8\hat{k}}{8}$
$= \frac{7}{4}\hat{i} + \frac{7}{4}\hat{j} + \hat{k}$
Distance of com from 2 kg mass:
$d = \sqrt{\left(\frac{7}{4}-1\right)^2 + \left(\frac{7}{4}-1\right)^2 + (1-1)^2}$
$d = \sqrt{\left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^2}$
$d = \frac{3\sqrt{2}}{4}$

(150) Answer : (1)
Solution:


These three forces will form a closed triangle and hence net force will be zero

(151) Answer : (3)
Solution:
Escape speed from earth's surface, $v_e = \sqrt{\frac{2GM}{R}}$. Let *h* is the maximum height reached by the rocket, then applying energy conservation.
$\frac{-GMm}{R} + \frac{1}{2}m\left(\frac{v_e}{2}\right)^2 = -\frac{GMm}{R+h}$
$\frac{-GMm}{R} + \frac{1}{2}m \times \frac{2GM}{4R} = -\frac{GMm}{R+h}$
$\frac{-1}{R} + \frac{1}{4R} = \frac{-1}{R+h} \Rightarrow \frac{-3}{4R} = \frac{-1}{R+h}$
$R+h = \frac{4R}{3} \Rightarrow h = \frac{R}{3}$ from earth's surface or $\frac{4R}{3}$ from Earth's centre.

(152) Answer : (3)
Solution:
Number of photoelectrons emitted depends on the photon intensity of light

(153) Answer : (1)
Solution:
$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mVq}} \Rightarrow \lambda \propto \frac{1}{\sqrt{V}}$
New voltage = $150 + \left(150 \times \frac{300}{100}\right) = 600\ \text{kV}$
$\frac{\lambda}{\lambda'} = \sqrt{\frac{600}{150}} \Rightarrow \frac{\lambda}{\lambda'} = 2 \Rightarrow \lambda' = \frac{\lambda}{2}$

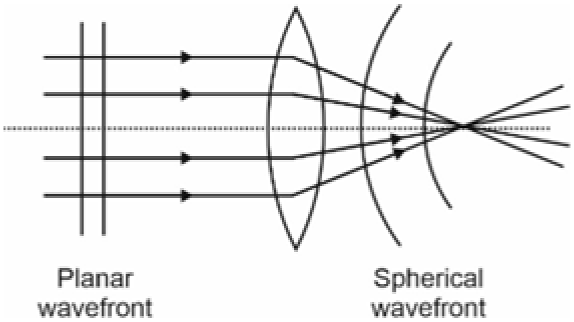
(154) Answer : (4)
Solution:
In SHM, acceleration and displacement have π phase difference.

$a = -\omega^2 x$

Hence, graph of $x - t$ will be negative of graph of $a - t$.

(155) Answer : (2)

Solution:



(156) Answer : (2)

Solution:

Given, f = 500 Hz, v = 330 m/s

$v = \lambda \times f \ \Rightarrow \ 330 = \lambda \times 500$

$\lambda = \frac{33}{50}\mathrm{m}$

$\Delta\phi = \frac{2\pi}{\lambda}\Delta x \ \Rightarrow \ \frac{\pi}{6} = \frac{2\pi}{\lambda}\Delta x$

$\Delta x = \frac{\lambda}{12} = \frac{33}{50} \times \frac{1}{12}\mathrm{m} = 0.055 \ \mathrm{m}$

(157) Answer : (3)

Solution:

Fundamental mode of a closed pipe:

$f = \frac{v}{4L}$

$f_1 - f_2 = 10$ …(i)

$\frac{f_1}{f_2} = \frac{L_2}{L_1} = \frac{26}{25}$ …(ii)

Solving equations (i) and (ii), we get

$f_1 = 260$ Hz, $f_2 = 250$ Hz

(158) Answer : (2)

Solution:

For damped oscillator,

$A = A_0 e^{-bt/2m}$

$\frac{A_0}{2} = A_0 e^{-bT/2m} \Rightarrow \ e^{-bT/2m} = \frac{1}{2}$ …(i)

$\frac{A_0}{8} = A_0 e^{-bt/2m}$

$\left(\frac{1}{2}\right)^3 = e^{-bt/2m} \Rightarrow \left(e^{-bT/2m}\right)^3 = e^{-bt/2m}$

On comparing, we get $t = 3T$

(159) Answer : (2)

Solution:

Capacitance of a capacitor is charge independent. When two charged capacitors are connected together, then usually energy is dissipated during charge distribution.

(160) Answer : (2)

Solution:

Work done by external agent = Change in potential energy.

When the charges were at infinity, $U_i = 0$

$U_f = \frac{2Kq^2}{r} + \frac{Kq^2}{2r} = \frac{5Kq^2}{2r}$

$W = U_f - U_i = \frac{5}{2} \times \frac{1}{4\pi\varepsilon_0} \times \frac{q^2}{r} - 0$

$W = \frac{5q^2}{8\pi\varepsilon_0 r}$

(161) Answer : (3)

Solution:

Electric dipole moment : $P = qd$

$P = 2.5 \times 10^{-7} \times 30 \times 10^{-2}$

$P = 7.5 \times 10^{-8}$ C m

(162) **Answer :** (3)

Solution:

$l_A - l_0 = \frac{T_A \times l_0}{A \times Y}$ and $l_B - l_0 = \frac{T_B \times l_0}{A \times Y}$

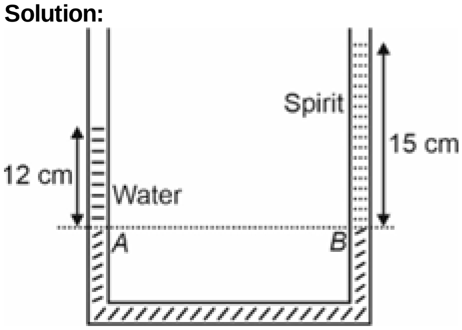
Y is same then

$= \frac{l_A - l_0}{T_A} = \frac{l_B - l_0}{T_B} = T_B l_A - T_A l_B = l_0 \, [T_B - T_A]$

$= l_0 = \frac{T_B l_A - T_A l_B}{T_B - T_A}$

(163) **Answer :** (1)

Solution:



$P_A = P_B \Rightarrow \rho_w \times g \times 12 = \rho_s \times g \times 15$

$\frac{\rho_s}{\rho_w} = \frac{12}{15} \Rightarrow$ specific gravity = $\frac{4}{5} = 0.8$

(164) **Answer :** (4)

Solution:

Isothermal – Temperature constant

Isobaric – Pressure constant

Isochoric – Volume constant

Adiabatic – No heat flow between system and surroundings

(165) **Answer :** (2)

Solution:

Each vibrational modes has energy equal to $K_B T$

(166) **Answer :** (2)

Solution:

A to B is an isothermal process,

hence $U_B = U_A = 200$ J

(167) **Answer :** (2)

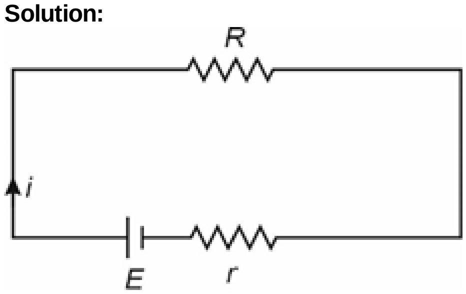
Solution:

$I = neAv_d \Rightarrow v_d = \frac{I}{neA}$

$\nu_d = \frac{1.5}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-7}} = 1.1 \times 10^{-3} \mathrm{m/s}$

(168) **Answer :** (4)

Solution:

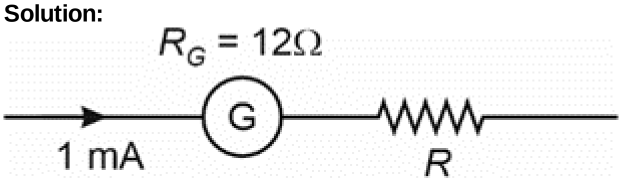


$i = \frac{E}{R+r}$

For maximum current ⇒ $R = 0$

(169) Answer : (2)
Solution:
Elementary particles such as an electron or a proton carry an intrinsic magnetic moment, not accounted by circulating currents

(170) Answer : (2)
Solution:



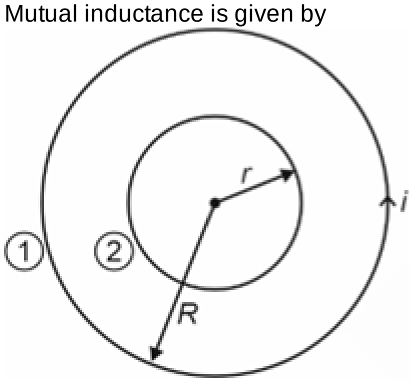
$V = i\,(R + R_G)$

$10 = 1 \times 10^{-3}\,(R + 12)$

R + 12 = 10^4 ⇒ R = 1000 –12 = 9988 Ω

(171) Answer : (2)
Solution:
Magnetic flux is outward through the circular coil and it starts decreasing when it is stretched as given in the question, hence according to Lenz's law to oppose the flux change current will flow anticlockwise.

(172) Answer : (2)
Solution:
Mutual inductance is given by



$M = \frac{\phi}{i}$

$M = \frac{\phi_2}{i_1}$ …(i)

$\phi_2 = B_1 A_2 = \frac{\mu_0 i \times \pi r^2}{2R}$ …(ii)

$M = \frac{\mu_0 \pi r^2}{2R}$

(173) Answer : (2)
Solution:
Given $|X_L - X_c| = 100\ \Omega,\quad R = 100\,\Omega$

$z = \sqrt{R^2 + (X_L - X_c)^2} = \sqrt{100^2 + 100^2} = 100\sqrt{2}\,\Omega$

$i = \frac{V}{z} = \frac{220}{100\sqrt{2}} = \frac{110 \times 2}{100\sqrt{2}} = 1.1\sqrt{2}\ \mathrm{A}$

(174) Answer : (4)
Solution:

$\underset{0^\circ C}{\text{Ice}} \overset{Q_1}{\rightarrow} \underset{0^\circ C}{\text{Water}} \overset{Q_2}{\rightarrow} \underset{\text{T}^\circ C}{\text{Water}}$

$Q_1 = mL$ ⇒ Q_1 = 80×15 = 1200 cal …(i)

$Q_2 = ms\Delta T$ ⇒ Q_2 = 15×1×T ⇒ Q_2 = 15 T …(ii)

$\underset{50^\circ C}{\text{Water}} \overset{Q_3}{\rightarrow} \underset{\text{T}^\circ C}{\text{Water}}$

(175) Answer : (3)

Solution:
For an isotropic solid ⇒ γ = 3α

(176) Answer : (1)
Solution:
$\mu_0\varepsilon_0 = \frac{B_0}{E_0 c}$ is the conclusion that can be drawn when electromagnetic wave satisfies the Ampere's law.

(177) Answer : (2)
Solution:
Δ = Shift in depth = $h\left(1-\frac{1}{\mu}\right)$

3.46 = 17.3 $\left(1-\frac{1}{\mu}\right)$

$\frac{3.46}{17.3} = \left(1-\frac{1}{\mu}\right) \therefore \ \frac{1}{5} = 1-\frac{1}{\mu}$

$\frac{1}{\mu} = 1-\frac{1}{5} = \frac{4}{5}$

$\mu = \frac{5}{4} = 1.25$

(178) Answer : (1)
Solution:
Given A = 60º, $\delta_m = 60^\circ$

$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60^\circ+60^\circ}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)} = \frac{\sin 60^\circ}{\sin 30^\circ}$

$\mu = \frac{\sqrt{3}/2}{1/2} = \sqrt{3} = 1.732$

(179) Answer : (2)
Solution:

Simple microscope	–	*Virtual image, erect and enlarged*
Compound microscope	–	*Magnified image, inverted and virtual*
Astronomical telescope	–	*Virtual, inverted and high resolution*
Terrestrial telescope	–	*Image virtual, erect and high resolution*

(180) Answer : (3)
Solution:
Difference of two successive resonance lengths = $\frac{\lambda}{2}$

$\therefore \quad 80-25 = \frac{\lambda}{2} \Rightarrow l = 110\,\text{cm}$

f = 340 Hz, *v* = ?

$v = \lambda \times f \Rightarrow v = \frac{110}{100} \times 340$

v = 374 m/s